

Executive Summary

Causal Impact of the NSW Job Training Program on Earnings

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We estimate that the NSW training program increases annual post-program earnings by \$1,541 per participant (95% CI: [\$229, \$2,853]).

Situation

The National Supported Work (NSW) Demonstration was a federally-funded U.S. job training program targeting disadvantaged workers in the 1970s. Policymakers today must decide whether to expand similar programs and at what scale. A defensible answer requires isolating the causal effect of training on earnings, not merely the correlation.

Complication

A simple comparison of earnings between trained and untrained workers yields a naive estimate of $-\$8,498$ — implying training reduces earnings. This is driven by severe selection bias: NSW participants were chosen precisely because they had the weakest labor market histories (mean pre-treatment earnings of $\$2,096$ vs. $\$14,017$ for the CPS comparison group). A prediction-only approach would confirm this misleading correlation. Our predictive Random Forest model achieves $R^2 = 0.421$ but assigns the treatment variable a feature importance of only 0.0025, demonstrating that prediction cannot isolate the causal effect.

Resolution

Using Double Machine Learning (DML) with Random Forest nuisance models and discrete-treatment propensity score estimation, we recover an average treatment effect of $\$1,541$ (95% CI: $[\$229, \$2,853]$). This aligns closely with the experimental RCT benchmark of $\$1,794$, providing strong validation of the causal identification. A robustness check using Gradient Boosting nuisance models yields $\$365$ (95% CI: $[-\$814, \$1,543]$) — the point estimate is lower but directionally consistent, with confidence intervals that overlap substantially with the primary specification.

Recommendation

Continue investment in targeted job training programs for disadvantaged populations. The directional conclusion — that training produces positive earnings gains — is robust across model specifications. **The return on investment is conditionally positive:** the $\$1,541$ annual earnings gain against the $\sim\$12,500$ per-participant program cost implies a payback period of approximately 8 years, so ROI depends on whether treatment effects persist that long. Under the 5-year persistence assumption used as the dashboard default, net benefit remains negative; under sustained-effect assumptions, the program delivers positive social returns. We further recommend commissioning a modern randomized trial to validate this finding for contemporary labor market conditions.

Key Assumption & Caveat

The estimate relies on **conditional independence** (unconfoundedness): given observed covariates (age, education, race, marital status, pre-treatment earnings), treatment assignment is independent of potential outcomes. This assumption fails if unobserved factors — such as individual motivation, mental health, or social networks — jointly drive both program participation and future earnings. The close alignment with the RCT benchmark provides partial reassurance, but the observational estimate cannot claim the same internal validity as a randomized experiment.